

Probability density function for continuous variable

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Continuous random variable

Density function

The probability density function is non negative everywhere, and its integral over the entire space is equal to one. That is for absolutely continuous univariate distribution

so

- 1 $f(x) \geq 0$, for all $x \in \mathbb{R}$.
- 2 $\int_{-\infty}^{\infty} f(x)dx = 1$.
- 3 $P(a < X < b) = \int_a^b f(x)dx$

Real-World IT Data Is Continuous

You don't get “3 latencies.” You get:

- Network latency: **42.7 ms**
- Server response: **0.341 seconds**
- CPU temperature: **68.25°C**
- Download speed: **12.8 Mbps**

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We model them using **continuous random variables**.

But... You Can't Say "*Probability* = *X*"

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Why? There are infinitely many possible values!

So instead of probabilities at single points, we ask:

$$P(90 \leq \text{latency} \leq 110) = ?$$

This is where the pdf comes in.

The pdf: A “Probability Map”

The **probability density function (pdf)**, $f(x)$, tells us **where values are likely**.

- High $f(x) \rightarrow$ values near x are common.
- Low $f(x) \rightarrow$ values near x are rare.

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The **area under the curve** gives real probabilities:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

(Think: “How much ink is under the curve from a to b ?”)

Why Should You Care?

With PDFs, you can:

- Set **realistic SLAs**: “95% of requests finish within 1 second.”
- Tune **cloud auto-scaling** based on CPU usage patterns.
- Detect **anomalies**: “This response time is in the worst 1%!”
- Understand **AI uncertainty**: “How confident is this prediction?”

Without pdfs, you only see the **average** and miss **delays, spikes, and risks**.

Example

Suppose that the error in the reaction temperature, in C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$.

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Solution

- (a) Obviously, $f(x) \geq 0$. $\int_{-\infty}^{\infty} f(x)dx = \int_{-1}^2 \frac{x^2}{3}dx = \frac{x^3}{9} \Big|_{-1}^2 = 1$
- (b) $P(0 < X \leq 1) = \int_0^1 \frac{x^2}{3}dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}$

Expectation & Variance

Let X be a continuous random variable with probability density function (pdf) $f(x)$.

Expected value

$$\mathbb{E}[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

Variance

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \left(\int_{-\infty}^{\infty} x^2 f(x) dx \right) - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Standard deviation

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{\mathbb{E}[X^2] - (\mathbb{E}[X])^2}$$

Example

A cloud-based web server handles user requests. The response time (in seconds), denoted by the continuous random variable X . Its probability density function (pdf) is:

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Tasks

- 1 Find the expected (average) response time.
- 2 Compute the variance of the response time.
- 3 Determine the standard deviation.
- 4 Interpret the results for system performance.

1. Expected Value $\mathbb{E}[X]$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) dx = \int_0^{\infty} x \cdot 2e^{-2x} dx$$

Use integration by parts: let

$$u = x \Rightarrow du = dx, \quad dv = 2e^{-2x} dx \Rightarrow v = -e^{-2x}$$

Then:

$$\int_0^{\infty} x \cdot 2e^{-2x} dx = \left[-xe^{-2x} \right]_0^{\infty} + \int_0^{\infty} e^{-2x} dx$$

Evaluate the boundary term: $\lim_{x \rightarrow \infty} (-xe^{-2x}) = 0$ (exponential decays faster than linear growth), and at $x = 0$: $-0 \cdot e^0 = 0$. So the first term is 0. Now compute the remaining integral:

$$\int_0^{\infty} e^{-2x} dx = \left[-\frac{1}{2}e^{-2x} \right]_0^{\infty} = 0 - \left(-\frac{1}{2} \right) = \frac{1}{2}$$

Thus,

$$\boxed{\mathbb{E}[X] = \frac{1}{2} \text{ seconds}}$$

Variance and standard deviation

We need $\mathbb{E}[X^2]$ to find the variance:

$$\mathbb{E}[X^2] = \int_0^{\infty} x^2 \cdot 2e^{-2x} dx$$

Apply integration by parts **twice**. First, let:

$$u = x^2 \Rightarrow du = 2x dx, \quad dv = 2e^{-2x} dx \Rightarrow v = -e^{-2x}$$

Then:

$$\int_0^{\infty} x^2 \cdot 2e^{-2x} dx = \left[-x^2 e^{-2x} \right]_0^{\infty} + \int_0^{\infty} 2xe^{-2x} dx$$

Cont...

The boundary term is 0 (same reasoning as before). So we are left with:

$$\int_0^{\infty} 2xe^{-2x} dx$$

But this is exactly the integral we computed for $\mathbb{E}[X]$, which equals $\frac{1}{2}$. Therefore:

$$\mathbb{E}[X^2] = \frac{1}{2}$$

Now use the computational formula:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Exercises

Problem 1

The weekly demand for a drinking-water product, in thousands of litres, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x - 1), & 1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate the expected value (mean) and variance.

Problem 2

CPU Utilization in a Virtual Machine

The CPU usage fraction $X \in [0, 1]$ of a virtual machine has pdf:

$$f(x) = 6x(1 - x), \quad 0 \leq x \leq 1$$

- 1 Show that $\int_0^1 f(x) dx = 1$.
- 2 Compute $\mathbb{E}[X]$.
- 3 Compute $\text{Var}(X)$ and σ .
- 4 If autoscaling triggers at $X = 0.8$, how variable is the load near this threshold?